Parallel and Distributed Systems

Exam Preparation

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Copyright & Acknowledgments

Chapter 2 + Exercises 2,3: Theory

- Problem size
- Runtime components
- Parallelization overhead
- Speedup
- Linear, superlinear, real speedup
- Amdahl’s Law incl. variables, extreme values
- Karp-Flatt value
- Gustafson’s Law incl. variables, extreme values
- Amdahl vs. Gustafson
- Inherent parallelism
- Data parallelism, task parallelism, pipeline parallelism

Chapter 2 + Exercises 2,3: Theory (contd.)

Not relevant:

- Deriving Amdahl’s and Gustafson’s Law
- Costs and Efficiency Metrics
- Divide and conquer
- Foster’s Methodology

Ch. 3 + Exercise 4: Process-Based Parallelism

- Processes from OS perspective
- Process (memory) components
- Process creation
- Linux process creation in detail
- Copy-on-write
- Process termination
- Process restarts
- Daemon processes
- Python: Process creation with arguments, start, join

Ch. 4 + Ex. 5,6,9: Inter-Proc. Communication

- Why IPC?
- Synchronous vs. asynchronous communication
- Blocking vs. non-blocking IPC
- Polling, Callback Functions, Futures
- Shared memory
  - How does it work?
  - Steps needed to use shared memory
  - Python: Multiprocessing Value and Array, General-Purpose Shared Memory
  - (File-Based) Memory Mapping vs. shared memory
  - mmap vs. read/write to file

Ch. 4 + Ex. 5,6,9: Inter-Proc. Communication (contd.)

- Message passing
  - How does it work?
  - Pipes
  - Message Queues
  - (File-Based) Memory Mapping vs. shared memory
  - mmap vs. read/write to file
- IPC Design Decisions
- Shared Memory vs. Message Passing
- Design Patterns
- Not relevant
  - Multiprocessing Manager
  - MPI

Ch. 5 + Exercise 7: Thread-Based Parallelism

- Threads from OS perspective
- Thread (memory) components
- Thread-Local Storage
- Threads vs. Processes
- When to use threads?
- Python threads
- Global Interpreter Lock
- Python: Thread creation with arguments, start, join, restarts, daemon threads
- Not relevant
  - Thread creation and termination

Ch. 6 + Ex. 8,9: Synchronization Mechanisms

- Critical sections
- Race conditions
- Atomic operations
- Locks
- Deadlocks
- Reentrant Locks
- Condition Variables
- Barriers
- Semaphores

Ch. 6 + Ex. 8,9: Synchronization Mechanisms (contd.)

- Queues for Thread Communication
- Lock-Free Data Structures
- Process vs. Thread Safety

- Not relevant
  - Events

Chapter 7: Asynchronous Programming

- What is it?
- Event Loop
- Coroutines
- Tasks
- Futures
- Pros and Cons

- Not relevant
  - Python code

Chapter 8: Introduction to Distributed Systems

- Challenges in Distributed Systems
- Definition Distributed System
- Middleware
- Client-Server Architecture
- Cluster and Cloud Computing

- Not relevant
  - All other distributed system architectures

Chapter 9 + Exercise 10: Sockets

- What are sockets?
- Steps in working with sockets
- Socket domains and addresses
- UDP
- Use Cases of UDP
- UDP Client and Server Implementation
- Multicast
- TCP
- Use Cases of TCP
- TCP Client and Server Implementation
- Serving multiple clients

Chapter 10: Remote Communication Models

- Motivation of RPCs
- Stubs and Skeletons
- Interface Definition Language
- Registry
- RPC Parameters and Return Values
- Error Handling

- Not relevant
  - gRPC
  - REST

Chapter 11: Hardware

- Von Neumann Architecture
- Von Neumann Bottleneck
- Crossbar Switches and Mulistage Networks
- Instruction-Level Parallelism
- In-order vs. out-of-order execution
- Thread-Level Parallelism
- Simultaneous Multithreading
- Multicore Processors
- Multicore vs. manycore
- Tile Architecture
- CPUs vs. GPUs

Chapter 11: Hardware (contd.)

- Not relevant
  - Synchonization Mechanisms and the CPU

Exam Procedure

- There are pre-generated question sheets with an equal number of questions
- After entering you pick one out of three sheets randomly (without seeing the questions before) – reduces examiner’s bias
- Each sheet in logically structured in three parts
  - Part 1: 5 true-false statements, explain your decision (Caution: Randomly assembled, may happen that all are true or all are false)
  - Part 2: Knowledge questions
  - Part 3: Knowledge questions + use case (code example or similar, see exercises)